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Inventor(s)

HONG; CHAOZHEN et al.

SEMICONDUCTOR STRUCTURE AND METHOD OF FABRICATING THE SAME

Abstract

The present disclosure provides a semiconductor structure and a fabricating method thereof, including a pad array. The pad array includes a plurality of first pads, a first marginal pad, a second marginal pad, a plurality of second pads, and a plurality of third pads. The first pads are separately disposed. The first marginal pad is disposed at one side of the pad array, and includes a plurality of first branch pads. The second marginal pad is disposed at another side of the pad array being opposite to the first marginal pad, and includes a plurality of second branch pads. The second pads are separately disposed between the first branch pads. The third pads are separately disposed between the second branch pads, wherein a length of the second pads is smaller than a length of the third pads.

Inventors: HONG; CHAOZHEN (Quanzhou City, CN), Tung; Yu-Cheng (Quanzhou City, CN), Wong; Wen-Yi (Quanzhou City, CN)

Applicant: Fujian Jinhua Integrated Circuit Co., Ltd. (Quanzhou City, CN)

Family ID: 1000007854125

Assignee: Fujian Jinhua Integrated Circuit Co., Ltd. (Quanzhou City, CN)

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present disclosure generally relates to a semiconductor structure and a method of fabricating the same, and more particularly, to a semiconductor structure including pads and a method of fabricating the same.

2. Description of the Prior Art

[0002] With the trend of miniaturization of various electronic products, the design of semiconductor devices must also meet the requirements of high integration and high density. Under the current mainstream of development trend, dynamic random access memories (DRAMs) having recessed gate structures have gradually replaced the DRAMs having only planar gate structures due to longer carrier channel length for the same semiconductor substrate so as to reduce current leakage of capacitor structures. In general, a DRAM cell with a recessed gate structure includes a transistor component and a charge storage device to receive voltage signals from bit lines and word lines. However, due to the limitations of current processing technologies, there are still many defects in currently available DRAM cells with recessed gate structures, which need to be further improved to effectively improve the performance and reliability of related memory devices.

SUMMARY OF THE INVENTION

[0003] An object of the present disclosure is to provide a semiconductor structure and a method of fabricating the same, in which peripheral pads with various lengths are arranged at two sides of the pad array, for improving the possible structural defects on the semiconductor structure which may be caused by the continuous increased density of the DRAM cell.

[0004] In order to achieve the above object, one embodiment of the present disclosure provides a semiconductor structure including a storage node pad array. The storage node pad array includes a plurality of first pads, a first marginal pad, a second marginal pad, a plurality of second pads, and a plurality of third pads. The first pads are separately disposed in a first direction and in a second direction. The first marginal pad is disposed at one side of the storage node pad array, and includes a plurality of first branch pads extending in the first direction. The second marginal pad is disposed at another side of the storage node pad array being opposite to the first marginal pad, and includes a plurality of second branch pads extending in the second direction. The second pads are separately disposed between the first branch pads in a third direction. The third pads are separately disposed between the second branch pads in the third direction, wherein a length of the second pads in the first direction is smaller than a length of the third pads in the first direction.

[0005] In order to achieve the above object, one embodiment of the present disclosure provides a semiconductor structure including a storage node pad array. The storage node pad array includes a plurality of first pads, a first marginal pad, and a second marginal pad. The first pads are separately arranged into a plurality of columns in a first direction, and a plurality of rows in a second direction. The first marginal pad is disposed at one side of the storage node pad array, and includes a plurality of first branch pads. The second marginal pad is disposed at another side of the storage node pad array being opposite to the first marginal pad, and includes a plurality of second branch pads. The first pads within each of the columns closest to the first marginal pad includes a first zigzag profile, the first pads within each of the columns closest to the second marginal pad includes a second zigzag profile, and the second zigzag profile is different from the first zigzag profile.

[0006] In order to achieve the above object, one embodiment of the present disclosure provides a fabricating method of a semiconductor structure including the following steps. A plug array is formed on a chip, and the plug array includes a plurality of plugs each including a plug-barycenter being defined thereon. A predicted storage node pad array is defined on the chip, wherein the predicted storage node pad array includes a plurality of predicted pad patterns, and each of the predicted pad patterns includes a predicted pad-barycenter being defined thereon. A plurality of pads is formed on the plugs through the predicted storage node pad array, wherein each of the pads overlaps each of the plugs and includes a pad-barycenter. The pads are adjusted through comparing positions of the plug-barycenter, the predicted pad-barycenter, and the pad-barycenter.

[0007] Overall speaking, According to the semiconductor structure and the fabricating method thereof of the present disclosure, peripheral pads with various lengths are arranged at two opposite sides of the SN pad array, with the SN pad array presenting in an asymmetric layout as a whole, to as to maintain the contacting area between the pads and the corresponding plugs within the cell region. Thus, the semiconductor structure of the present disclosure enables to gain an optimized structure and better functions.

[0008] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings are directed to provide a better understanding of the embodiments and are included as parts of the specification of the present disclosure. These drawings and descriptions are used to illustrate the principles of the embodiments. It should be noted that all drawings are schematic, and the relative dimensions and scales have been adjusted for the convenience of drawing. Identical or similar features in different embodiments are marked with identical symbols.

[0010] FIG. 1 to FIG. 4 are schematic diagrams illustrating a semiconductor structure according to one preferably embodiment of the present disclosure, wherein:

[0011] FIG. 1 illustrates a top view of a semiconductor structure;

[0012] FIG. 2 illustrates a cross-sectional view taken along cross lines A-A' and B-B';

[0013] FIG. 3 illustrates another top view of a semiconductor structure; and

[0014] FIG. 4 illustrates a partial enlarged view of FIG. 3.

[0015] FIG. 5 to FIG. 9 are schematic diagrams illustrating a fabricating method of a semiconductor structure according to one preferably embodiment of the present disclosure, wherein:

[0016] FIG. 5 illustrates a process flow of the fabricating method of the semiconductor structure;

[0017] FIG. 6 illustrates a top view of a semiconductor structure after forming a plug array, a predicted storage node pad array, and a pad array;

[0018] FIG. 7 illustrates a cross-sectional view of a semiconductor structure after forming a plug array, a predicted storage node pad array, and a pad array;

[0019] FIG. 8 illustrates a top view of a semiconductor structure after adjusting the pad array; and

[0020] FIG. 9 illustrates a cross-sectional view of a semiconductor structure after adjusting the pad array.

[0021] FIG. 10 illustrates another top view of a semiconductor structure after adjusting the pad array according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0022] For better understanding of the presented disclosure, preferred embodiments will be

described in detail. The preferred embodiments of the present disclosure are illustrated in the accompanying drawings with numbered elements. In addition, the technical features in different embodiments described in the following may be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.

[0023] Please refer to FIG. 1 to FIG. 4, which are schematic diagrams illustrating a semiconductor structure **10** according to a preferable embodiment of the present disclosure. Firstly, as shown in FIG. 1, the semiconductor structure **10** includes a storage node (SN) pad array **130** for example including a low-resistance metal material like aluminum (Al), titanium (Ti), copper (Cu) or tungsten (W), and preferable for tungsten, but not limited. The SN pad array **130** precisely includes a plurality of first pads **131**, a first marginal pad **133**, a second marginal pad **137**, a plurality of second pads **141** and a plurality of third pads **143**. The first pads **131** are separately from each other in a first direction **D1** and in a second direction **D2** which are interlaced and not perpendicular to each other. The first marginal pad **133** and the second marginal pad **137** are respectively extended in a third direction **D3**, at two opposite sides of the SN pad array **130**. The first marginal pad **133** disposed at one side of the SN pad array **130** further includes a plurality of first branch pads **135** in the first direction **D1**, and the second marginal pad **137** disposed at another side of the SN pad array **130** further includes a plurality of second branch pads **139** in the first direction **D1**. The second pads **141** are separately disposed between the first branch pads **135** in the third direction **D3**, and the third pads **143** are separately disposed between the second branch pads **139** in the third direction **D3**.

[0024] It is noted that each of the first pads **131** includes a same first length **S1** in the first direction **D1**, and each of the second pads **141** and the third pads **143** respectively include a second length **S2** and a third length **S3** different from the first length **S1** in the first direction **D1**. The second length **S2** of the second pads **141** is smaller than the third length **S3** of the third pads **143**. Accordingly, through arranging second pads **141** and third pads **143** in various lengths (the second length **S2** and the third length **S3**) at two opposite sides of the SN pad array **130**, the SN pad array **130** will therefore present in an asymmetric layout as being viewed from a top view as shown in FIG. 1, thereby improving the possible structural defects of the semiconductor structure **10** caused by the continuous increased density of the DRAM cell. Also, in another embodiment, each of the first branch pads **135** and each of the second branch pads **139** at two opposite sides of the SN pad array **130** respectively include a fourth length **S4** and a fifth length **S5** different from the first length **S1**, and the fourth length **S4** of the first branch pads **135** is smaller than the fifth length **S5** of the second branch pads **139**, thereby presenting in an asymmetric layout as well.

[0025] Further in view of FIG. 1, there is no first pads **131** being disposed between adjacent ones of the first branch pads **135** or the second pads **141** in the third direction **D3**, at one side of the SN pad array **130**, and there is one of the first pads **131** being disposed between adjacent ones of the third pads **143**, at another side of the SN pad array **130**. Also, the end-portions of each first branch pad **135** and each second pad **141** at one side of the SN pad array **130** are aligned with each other in the third direction **D3**, such that, the end-portions of each first branch pad **135** and each second pad **141** are located on the same cross line “A”, as shown in FIG. 1. On the other hand, the end-portions of each second branch pad **139** and each third pad **143** are alternately arranged in the third direction **D3**, with the end-portions of the second branch pads **139** being aligned with each other and located on the same cross line “b”, and being misaligned with the end-portions of the third pads **143**. In the preset embodiment, the first marginal pad **133**, the second marginal pad **137** are respectively disposed outside all of the first pads **131**, the second pads **141**, and the third pads **143**, so as to gain the protection on the overall arrangement of the first pads **131**. Those who skilled in the art should fully realize that the semiconductor structure **10** in another embodiment may further include at least one third marginal pad **145** in a fourth direction **D4**, with two ends of the third marginal pad **145** respectively connecting the first marginal pad **133** and the second marginal pad **137**, so that, the first marginal pad **133**, the second marginal pad **137**, and the third marginal pad **145** will therefore

around all of the first pads **131**, the second pads **141**, and the third pads **143**, to present in a rectangular shape or in other suitable shape, to achieve better protection.

[0026] As shown in FIG. 1 and FIG. 2, the semiconductor structure **10** further includes a substrate **100** and a plurality of word lines **120** disposed within the substrate **100**. The above-mentioned SN pad array **130** is disposed on the substrate **100**, and the substrate **100** for example includes a silicon substrate, a silicon-containing substrate, e.g., SiC or SiGe, a silicon-on-insulator (SOI) substrate, or a substrate formed of any other suitable material, but it is not limited thereto. In principle, the first pads **131** are disposed within a cell region **100A** having a relative greater integration, and the second pads **141**, the first branch pads **135**, the third pads **143**, and the second branch pads **139** disposed at two sides of the SN pad array **130** are disposed within a periphery region **100B** having a relative lower integration. In one embodiment, the periphery region **100B** is for example disposed at least one side of the cell region **100A**. Preferably, the periphery region **100B** is for example disposed around the cell region **100A**, but not limited thereto. Each of the word lines **120** separately disposed within the substrate **100**, and precisely includes a dielectric layer **122**, a gate dielectric layer **124**, a gate **126**, and a capping layer **128** covered on the gate **126**, as shown in FIG. 2. The top surface of the capping layer **128** is coplanar with the top surface of the substrate **100**, such that, each word line **120** may serve as a buried word line (BWL) of a semiconductor device, being isolated from other components over the substrate **100** through an insulating layer **102** disposed on the substrate **100**. In one embodiment, the insulating layer **102** preferably includes a composite structure including an oxide-nitride-oxide (ONO) structure, but not limited thereto.

[0027] Further in view of FIG. 1 and FIG. 2, the semiconductor structure **10** further includes a plug array **110** also disposed on the substrate **100**, and which includes a plurality of plugs **111**, **113** separately arranged in the third direction D3 and in the fourth direction D4 which are perpendicular to each other, in an array arrangement as shown in FIG. 1. The plugs **111**, **113** are respectively within the cell region **100A**, and within the periphery region **100B**, being alternately arranged with a plurality of isolation structures in the fourth direction D4, for isolating the adjacent ones of the plugs **111**, **113**. In one embodiment, the plugs **111** for example includes an epitaxial material like silicon, silicon phosphorus (SiP), silicon germanium (SiGe) or germanium (Ge), or a low-resistance metal material like aluminum, titanium, copper or tungsten, but not limited thereto. It is noted that, although the plug array **110** is arranged in a different manner as that of the SN pad array **130**, each of the plugs **111** is disposed below each of the first pads **131** in a direction being perpendicular to the substrate **100**, to physically contact each of the first pads **131**. Each of the plugs **113** is disposed below the second branch pads **139** or the third pads **143**, to physically contact a corresponding one of the second branch pads **139** or the third pads **143**. The plugs **111** disposed within the cell region **100A** are partially extended into the substrate **100** through its bottom thereof, to electrically connect a transistor (not shown in the drawings) disposed within the substrate **100**, and to electrically connect the first pads **131** through its top thereof, thereby being configured as storage node contacts (SNCs) of the semiconductor device **10** to further connect to a storage node (SN) being formed subsequently. The plugs **113** disposed within the periphery region **100B** are disposed over the insulating layer **102**, without contacting the substrate **100** to be served as dummy plugs. It is also noted that, one of the first plugs **131** closest to the second pads **141** or the first branch pads **135**, and the plug **111** corresponding thereto have a barycenter C1, C2 respectively, and a minimum distance g1 between the barycenter C1 and the barycenter C2 is greater than a minimum distance g2 between a barycenter C3 of anyone of other first pads **131** and a barycenter C4 of the plug **111** corresponding thereto. Also, a minimum distance g3 between a barycenter C5 of one of the first plugs **131** closest to the second branch pads **139** or the third pads **143** and a barycenter C6 of the plug **111** corresponding thereto is greater than the minimum distance g2, and preferably smaller than the minimum distance g1, but not limited thereto.

[0028] On the other hand, as shown in FIG. 3 and FIG. 4, the first pads **131** are separately arranged into a plurality of columns **130a** in the first direction D1, and into a plurality of rows **130b** in the

second direction D2, thereby presenting in another array arrangement. It is noted that, the first pads **131** within each column **130a**, being closest to the second marginal pad **137**, are arranged into a first zigzag profile Z1 in the third direction D3, and the first pads **131** within each row **130b**, being closest to the first marginal pad **133**, are arranged into a second zigzag profile Z2. The first zigzag profile Z1 includes an opening O2 in the third direction D3, the second zigzag profile Z2 includes an opening O1 in the third direction D3, and the opening O2 of the first zigzag profile Z1 is greater than the opening O1 of the second zigzag profile Z2. Also, the first zigzag profile Z1 includes a height H2 in the fourth direction D4, the second zigzag profile Z2 includes a height H1 in the fourth direction D4, and the height H2 is larger than the height H1. The height H1, H2 is namely the height difference between a tip and a bottom of each zigzag profile.

[0029] With these arrangements, the semiconductor structure **10** of the preferably embodiment enables to arrange peripheral pads (including the second pads **141**, the first branch pads **135**, the third pads **143** and the second branch pads **139**) at two opposite side of the SN pad array **130**, to present in an asymmetric layout as a whole. Accordingly, the minimum distances g1, g2, g3 between the barycenter C1, C3, C5 of each first pad **131** and the barycenter C2, C4, C6 of each plug **111** corresponding thereto will be maintained in a certain range, especially for the minimum distances g1 g3 between the barycenter C1, C5 of the first pad **131** closed to the above-mentioned peripheral pads (for example the second pads **141**, the third pads **143**, and the second branch pads **139**) and the barycenter C2, C6 of the plug **111** corresponding thereto. In this way, it is sufficient to avoid the distance between the barycenter of the first pad **131** and the barycenter of the plug **111** excessive enlarge due to the misalignment between the barycenter C1, C5 of the first pads **131** and the barycenter C2, C6 of the plug **111**, such that, the contact area between the first pads **131** and the plugs **111** are seriously affected thereby.

[0030] Thus, the semiconductor structure **10** of the present embodiment is allowable to obtain an optimized structure and better functions. People who has ordinary skill in the art would easily understand that various components such as transistor components, word line components and/or bit line components may be additionally arranged within the cell region **100A** of the substrate **100**, due to practical product requirements. For example, a plurality of capacitors (not shown in the drawings) electrically connected to the first pads **131** may be further disposed above the SN pad array **130**, thereby forming a dynamic random access memory (DRAM) device to achieve better device performance, but not limited thereto.

[0031] In order to make those having ordinary skills in the art easily understand the semiconductor device **10** according to the present disclosure, a method of fabricating the semiconductor device **10** according to the present disclosure will be further described as follows.

[0032] Please refer FIG. 5 to FIG. 9, which are schematic diagrams illustrating a fabricating method of a semiconductor structure **10** according to the preferably embodiment of the present disclosure, with FIG. 5 showing a process flow of the fabricating method of the semiconductor structure **10**, with FIGS. 6-9 respectively showing various processing steps of the fabricating method of the semiconductor structure **10**. Firstly, as shown in FIG. 5 and FIG. 6, a chip **200** is provided, and the plug array **110** is formed on the chip (as shown in step S1). The plug array **110** includes a plurality of the plugs **111**, and a plug-barycenter **111a** is defined on each of the plugs **111**.

[0033] Next, a mask layer (not shown in the drawings) is formed on the chip **200**, covering the plug array **110**, and a predicted SN pad array **210** is defined on the mask layer, with the predicted SN pad array **210** including a plurality of predicted pad patterns **211** (as shown in step S2). Then, a predicted pad-barycenter **211a** is defined on each of the predicted pad patterns **211**. The predicted pad patterns **211** respectively overlap the plugs **111** underneath. Those who skilled in the art should fully realize that only partial plugs **111** and partial predicted pad patterns **211** are shown in FIG. 6 of the present embodiment, in order to clearly illustrate the relation between the plugs **111** and the predicted pad patterns **211**, and the practical arrangement thereof are not limited thereto. The plugs

111 are for example arranged in the third direction **D3** and the fourth direction **D4** with are perpendicular with each other, to present in an array arrangement as shown in FIG. 1, and the predicted pad patterns **211** are for example arranged in the first direction **D1** and the second direction **D2** which are crossed to each other, to present in another array arrangement also shown in FIG. 1, but not limited thereto.

[0034] As shown in FIG. 5 and FIG. 7, a photolithography process is performed on the chip **200** through the predicted SN pad array **210**, to form a SN pad array **230** including a plurality of pads **231**, with the pads **231** respectively overlapping with the plugs **111** underneath. It is noted that, the formation of the pads **231** is accomplished by performing two self-aligned reverse patterning (SARP) processes, to simultaneously form the pads **231**, marginal pads **233**, **237** having a plurality of branch pads **235**, **239** in the first direction **D1**, and peripheral pads **241**, **243** alternately arranged with the branch pads **235**, **239** in the third direction **D3**, as shown in FIG. 6, but not limited thereto. Due to the loading effect of the etching process, at least a portion of the pads **231** has a reduced size or a misaligned position, in comparison with that of the predicted pad patterns **211**, and the overlaying rates between the pads **231** and the corresponding plugs **111** will be affected thereby. For example, one pad **231b** adjacent to the branch pads **235** or the peripheral pads **241** may slightly shift toward the plug-barycenter **111b** of the corresponding plug **111a**, in the first direction **D1** or in the second direction **D2**, and another one pad **231c** adjacent to the branch pads **239** or the peripheral pads **243** may slightly shift away from the plug-barycenter **111b** of the corresponding plug **111c**, in the first direction **D1** or in the second direction **D2**, such that, the overlaying rates between the pads **231b**, **231c** and the corresponding plugs **111a**, **111c** will be influenced thereby. Although the forming positions of the pads **231b**, **231c** are all misaligned with the original positions of the predicted pad patterns **211**, the pad **231b** and the corresponding plug **111b** are still overlapped with each other to maintain a certain overlaying rate, as shown in FIG. 6 and FIG. 6. On the other hand, the pad **231c** and the corresponding plug **111c** cannot maintain the certain overlaying rate.

[0035] Following these, the positions of the pads **230** are adjusted through comparing the positions of the plug-barycenter **111a**, the predicted pad-barycenter **211a**, and the pad-barycenter **231a** (as shown in step S4). Precisely speaking, the reducing size or the shift position is easily occurred on the pads **231b**, **231c** adjacent to the branch pads **235**, **239** or the peripheral pads **241**, **243**, and which may be previously evaluated by defining an angle θ_1 , θ_2 between the plug-barycenter **111a**, the predicted pad-barycenter **211a**, and the pad-barycenter **231a**, as shown in FIG. 6. The angle θ_2 between the pad-barycenter **231a** of the pad **231c**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111c** is greater than the angle θ_1 between the pad-barycenter **231a** of the pad **231b**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111b**. Through these performances, the fabricating method of the present embodiment is allowable to previously estimate the shifting degree of the pads **231** by determining the degree of the angles between the pad-barycenter **231a** of one pad **231**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111** (as shown in step S4).

[0036] For example, if the angle for example being the angle θ_1 as shown in FIG. 6, between the pad-barycenter **231a** of one pad **231**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111** is an acute angle being less than 90 degree, the pad **231** still overlaps the corresponding plug **111** in a relative higher overlaying rate, with only a slight shifting occurred, such as the pad **231b** as shown in FIG. 6 and FIG. 7. On the other hand, if the angle for example being the angle θ_2 as shown in FIG. 6, between the pad-barycenter **231a** of one pad **231**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111** is a blunt angle being greater than 90 degree, the pad **231** may be seriously shifted from the

original position, being overlapped with the corresponding plug **111** in a relative lower overlaying rate, or being not overlapped with the corresponding plug **111**. In this way, the overlaying rate between each pad **231** and the corresponding plug **111** may be previously estimated by determining the angle between the pad-barycenter **231a** of one pad **231**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111**, followed by determining to remain the pad **231** or to modify the pad **231**. In other words, when the angle between the pad-barycenter **231a** of one pad **231**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111** is less than 90 degrees, the pad **231** may be remained. Otherwise, when the angle between the pad-barycenter **231a** of one pad **231**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111** is greater than 90 degrees, the pad **231** is then modified thereby.

[0037] In one embodiment, modifying the pad **231** for example includes removing the pads **231** followed by reforming a new SN pad array **330**. Precisely speaking, as shown in FIG. **8**, the SN pad array **330** includes the pads **231** originally shown in FIG. **6**, and a plurality of pads **331** additionally formed at one side of the pad **231c** (having the angle greater than 90 degrees), with the pads **331** being aligned with each other in the third direction, and also being aligned with the pad **231c** as shown in FIG. **6** in the third direction D3. Through these performances, the possible etching defects caused by the above-mentioned loading effect will be adjusted by additionally forming the pads **331** at one side of the pad **231c**, thereby improving the forming positions of the pads **331** adjacent to the marginal pad **237**. Then, all the pads **231**, **331** are overlapped with the corresponding plugs underneath in a certain overlaying rate, as shown in FIG. **9**, such that, the semiconductor structure formed accordingly will therefore obtain an improved structure to gain a better reliability. In another embodiment, modifying the pad for example includes removing the pad **231**, for example the pad **231c** as shown in FIG. **6**, as well as the pads **231** aligned with the aforementioned pad **231** in the third direction D3, while the angle between the pad-barycenter **231a** of the pad **231**, the plug-barycenter **111a** of the corresponding plug **111**, and the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211** is greater than 90 degrees. Those who skilled in the art should fully understand that since the predicted pad patterns **211** of the predicted SN pad array **211** are arranged in the first direction D1 and the second direction D2 into the array arrangement, the pads **231** formed accordingly in the present embodiment will also present in the same array arrangement. Thus, it is preferably to remove a row of the pads **231** at the same time, as modifying the pad **231** that having the angle greater than 90 degrees. That is, a SN pad array **430** is then formed by removing all the pads **231c** adjacent to the branch pads **239** or the peripheral pads **243**, as shown in FIG. **10**, and all the pads **231** remained on the SN pad array **430** will therefore overlap the corresponding plugs **111** underneath in a certain overlaying rate. Then, the semiconductor structure formed thereby may therefore obtain the improved structure and better reliability.

[0038] According to the fabricating method of the present embodiment, the possible shifting between each pad **231** and the corresponding plug **111** may be previously estimated by determining the angle between the pad-barycenter **231a** of one pad **231b**, **231c**, the predicted pad-barycenter **211a** of the corresponding predicted pad pattern **211**, and the plug-barycenter **111a** of the corresponding plug **111**, followed by keeping the pad **231** or modifying the pad **231**. Also, since the above-mentioned loading effect of the etching is generally occurred on the pads **231** at one side of the SN pad array **230**, the pad-barycenter **231a** of the pads **231c** at one side of the SN pad array **230** may be seriously shifted from the plug-barycenter **111a** of the corresponding plug **111**, and the overlaying rate between the pads **231c** and the plugs **111** will be dramatically affected thereby. Thus, the pads **231c** only disposed at one side of the SN pad array **230** are modified, and the new SN pad array formed subsequently will thereof present in an asymmetric layout, such as the SN pad array **130** as shown in FIG. **1**. Through these performances, the fabricating method of the present

embodiment is allowable to form the semiconductor structure **10** with the optimized structure and functions, which enables to be applied on forming a DRAM device subsequently, to achieve better performances thereby.

[0039] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor structure, comprising: a storage node pad array, the storage node pad array comprising: a plurality of first pads separately disposed in a first direction and in a second direction; a first marginal pad disposed at one side of the storage node pad array, the first marginal pad comprising a plurality of first branch pads extending in the first direction; a second marginal pad disposed at another side of the storage node pad array being opposite to the first marginal pad, the second marginal pad comprising a plurality of second branch pads extending in the first direction; a plurality of second pads separately disposed between the first branch pads in a third direction; and a plurality of third pads separately disposed between the second branch pads in the third direction; wherein a length of the second pads in the first direction is smaller than a length of the third pads in the first direction.
2. The semiconductor structure according to claim 1, wherein one of the first pads is sandwiched between the second branch pads or between the third pads, and none of the first pads is sandwiched between the first branch pads or between the second pads.
3. The semiconductor structure according to claim 1, wherein the first pads comprise a first length in the first direction, the second pads comprise a second length in the first direction, and the third pad comprise a third length in the first direction, wherein the first length, the second length and the third length are all different from each other.
4. The semiconductor structure according to claim 1, wherein end-portions of each of the second pads and each of the first branch pads are aligned with each other in the third direction, and end-portions of each of the third pads and each of the second branch pads are partially misaligned with each other.
5. The semiconductor structure according to claim 1, further comprising: a plug array disposed below the storage node pad array, and the plug array comprising: a plurality of plugs disposed below the first pads, the second pads and the third pads, wherein each of the first pads, each of the second pads, and each of the third pad at least partially overlap a corresponding one of the plugs underneath, respectively.
6. The semiconductor structure according to claim 5, wherein a distance between a barycenter of one of the first pads being closest to the second pads, the first branch pads, the third pads or the second branch pads in the first direction, and a barycenter of one of the plugs corresponding thereto is greater than a distance between a barycenter of another one of the first pads and a barycenter of one of the plugs corresponding thereto.
7. A semiconductor structure, comprising: a storage node pad array, the storage node pad array comprising: a plurality of first pads separately arranged into a plurality of columns in a first direction, and arranged into a plurality of rows in a second direction; a first marginal pad disposed at one side of the storage node pad array, the first marginal pad comprising a plurality of first branch pads; and a second marginal pad disposed at another side of the storage node pad array being opposite to the first marginal pad, the second marginal pad comprising a plurality of second branch pads; wherein the first pads within each of the columns closest to the first marginal pad comprises a first zigzag profile, the first pads within each of the columns closest to the second marginal pad comprises a second zigzag profile, and the second zigzag profile is different from the

first zigzag profile.

8. The semiconductor structure according to claim 7, wherein an opening of the first zigzag profile is greater than an opening of the second zigzag profile.

9. The semiconductor structure according to claim 7, wherein a height of the first zigzag profile is greater than a height of the second zigzag profile.

10. The semiconductor structure according to claim 7, the storage node pad array further comprising: a plurality of second pads separately disposed between the first branch pads in a third direction; and a plurality of third pads separately disposed between the second branch pads in the third direction, wherein the first pads comprise a first length in the first direction, the second pads comprise a second length in the first direction, the third pads comprise a third length in the first direction, and the first length, the second length, and the third length are all different from each other.

11. The semiconductor structure according to claim 10, wherein one of the first pads is sandwiched between the second branch pads or between the third pads in the third direction, and none of the first pads is sandwiched between the first branch pads or between the second pads in the third direction.

12. The semiconductor structure according to claim 10, further comprising: a plug array comprising: a plurality of plugs disposed below the first pads, the second pads and the third pads, wherein each of the first pads, each of the second pads, and each of the third pad at least partially overlap a corresponding one of the plugs underneath, respectively.

13. The semiconductor structure according to claim 12, wherein a distance between a barycenter of one of the first pads being closest to the second pads, the first branch pads, the third pads or the second branch pads in the first direction, and a barycenter of one of the plugs corresponding thereto is greater than a distance between a barycenter of another one of the first pads and a barycenter of one of the plugs corresponding thereto.

14. A fabricating method of a semiconductor structure, comprising: forming a plug array on a chip, the plug array comprising a plurality of plugs, and each of the plugs comprising a plug-barycenter being defined thereon; defining a predicted storage node pad array on the chip, wherein the predicted storage node pad array comprises a plurality of predicted pad patterns, and each of the predicted pad patterns comprises a predicted pad-barycenter being defined thereon; forming a plurality of pads on the plugs through the predicted storage node pad array, each of the pads overlapping each of the plugs and comprising a pad-barycenter; and adjusting the pads through comparing positions of the plug-barycenter, the predicted pad-barycenter, and the pad-barycenter.

15. The semiconductor structure according to claim 14, adjusting the pads further comprising: defining an angle between the pad-barycenter of one of the pads, and the plug-barycenter and the predicted pad-barycenter corresponding thereto; and modifying the one of the pads or keeping the one of the pads through the angle.

16. The semiconductor structure according to claim 15, further comprising: forming a storage node pad array, the storage node pad array comprising: a plurality of first pads separately arranged into a plurality of columns in a first direction, and a plurality of rows in a second direction; a plurality of second pads arranged at a first side of the first pads; and a plurality of third pads arranged at a second side of the first pads, wherein the first pads comprise a first length in the first direction, the second pads comprise a second length in the first direction, the third pads comprise a third length in the first direction, and the first length, the second length, and the third length are all different from each other.

17. The semiconductor structure according to claim 16, further comprising: keeping the one of the pads while the angle is less than 90 degrees.

18. The semiconductor structure according to claim 16, further comprising: modify the one of the pads while the angle is greater than 90 degrees.

19. The semiconductor structure according to claim 18, modifying the pad further comprising:

additionally forming a column of the first pads being aligned with each other in the third direction, on the plugs.

20. The semiconductor structure according to claim 18, modifying the pad further comprising: removing the first pads closest to the second pads or the third pads in the first direction.
